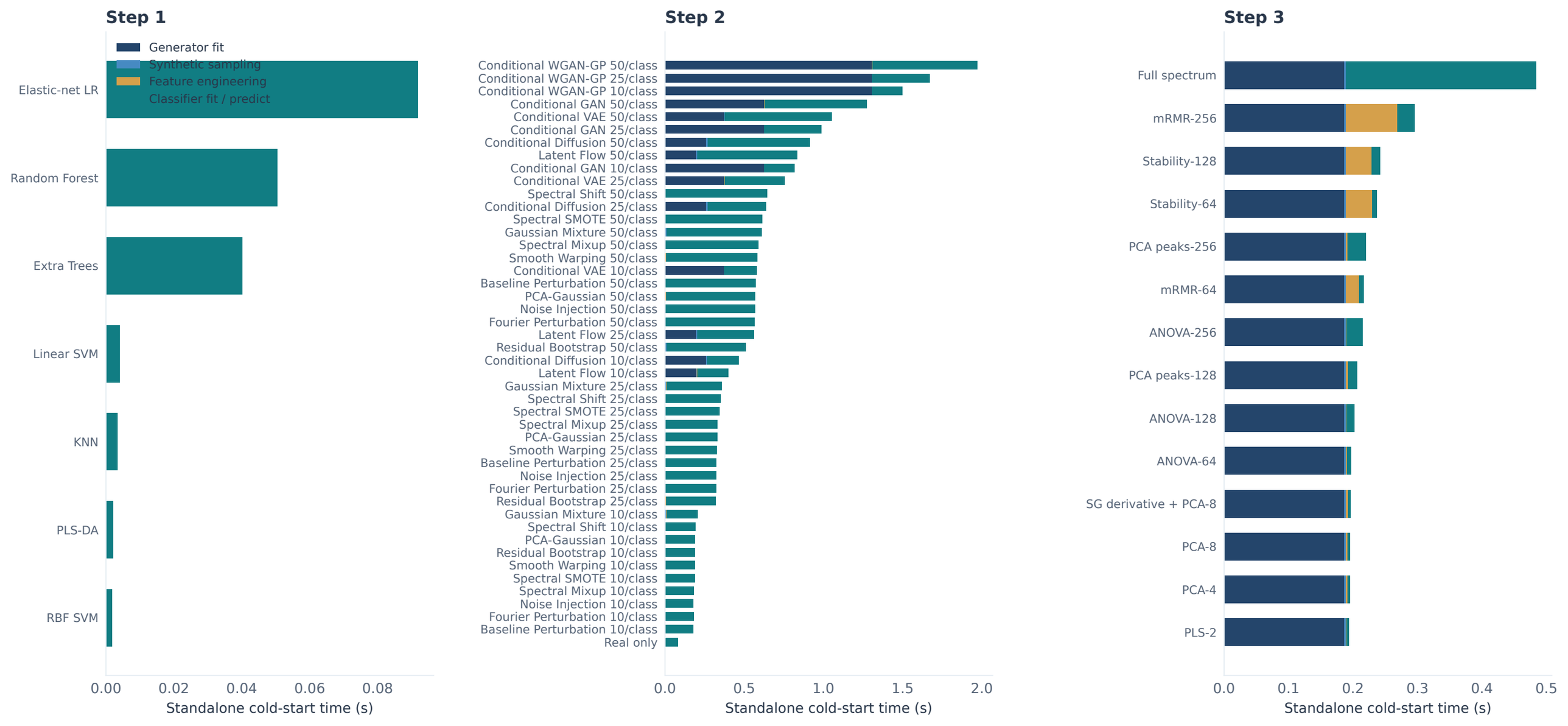
Step 1
Generator fit
Synthetic sampling
Feature engineering
Classifier fit / predict
Elastic-net LR
Random Forest
Extra Trees
Linear SVM
KNN
PLS-DA
RBF SVM
0.00
0.02
0.04
0.06
0.08
Standalone cold-start time (s)
Step 2
Conditional WGAN-GP 50/class
Conditional WGAN-GP 25/class
Conditional WGAN-GP 10/class
Conditional GAN 50/class
Conditional VAE 50/class
Conditional GAN 25/class
Conditional Diffusion 50/class
Latent Flow 50/class
Conditional GAN 10/class
Conditional VAE 25/class
Spectral Shift 50/class
Conditional Diffusion 25/class
Spectral SMOTE 50/class
Gaussian Mixture 50/class
Spectral Mixup 50/class
Smooth Warping 50/class
Conditional VAE 10/class
Baseline Perturbation 50/class
PCA-Gaussian 50/class
Noise Injection 50/class
Fourier Perturbation 50/class
Latent Flow 25/class
Residual Bootstrap 50/class
Conditional Diffusion 10/class
Latent Flow 10/class
Gaussian Mixture 25/class
Spectral Shift 25/class
Spectral SMOTE 25/class
Spectral Mixup 25/class
PCA-Gaussian 25/class
Smooth Warping 25/class
Baseline Perturbation 25/class
Noise Injection 25/class
Fourier Perturbation 25/class
Residual Bootstrap 25/class
Gaussian Mixture 10/class
Spectral Shift 10/class
PCA-Gaussian 10/class
Residual Bootstrap 10/class
Smooth Warping 10/class
Spectral SMOTE 10/class
Spectral Mixup 10/class
Noise Injection 10/class
Fourier Perturbation 10/class
Baseline Perturbation 10/class
Real only
0.0
0.5
1.0
1.5
2.0
Standalone cold-start time (s)
Step 3
Full spectrum
mRMR-256
Stability-128
Stability-64
PCA peaks-256
mRMR-64
ANOVA-256
PCA peaks-128
ANOVA-128
ANOVA-64
SG derivative + PCA-8
PCA-8
PCA-4
PLS-2
0.0
0.1
0.2
0.3
0.4
0.5
Standalone cold-start time (s)